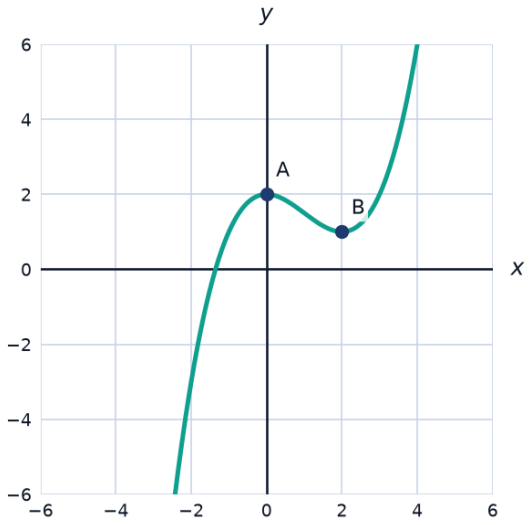
The Chain Rule & Directional Derivatives

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

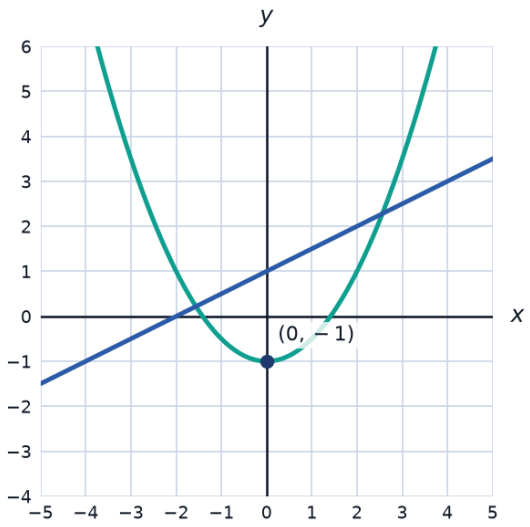
1 **[Foundational]** • **Target: Multivariable Chain Rule**

Find the directional derivative of $f(x, y) = x^2y - 3y^2$ at the point $P(1, 2)$ in the direction of the vector $\mathbf{v} = \langle 3, 4 \rangle$.



2 **[Foundational]** • **Target: The Gradient Vector**

Consider the function $z = f(x, y) = x^2 + y^2$. Let $x = 2t$ and $y = 3t$. Use the Multivariable Chain Rule to find $\frac{dz}{dt}$ when $t = 1$.



3 **[Foundational]** • **Target: Directional Derivatives**

For $f(x, y) = -12x - 35y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

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4 [Foundational] • Target: Direction of Maximum Increase & Steepest Ascent

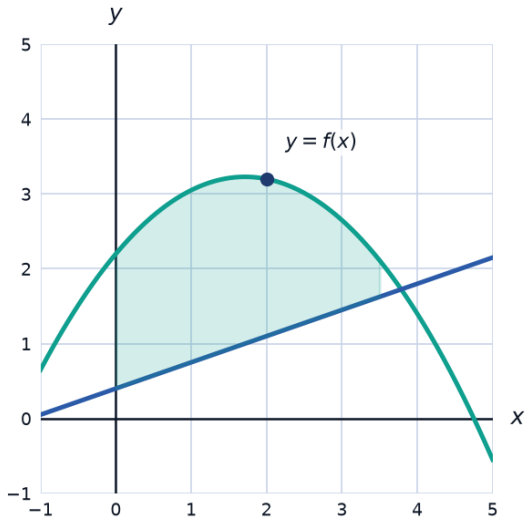
For $f(x, y) = -20x + 21y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

5 [Foundational] • Target: Multivariable Chain Rule

Prove that the 3D vector field $\mathbf{F} = \langle 2xy + z^2,\ x^2 + 2yz,\ 2xz + y^2 \rangle$ is conservative on $\mathbb{R}^3$, construct a potential function f satisfying $ablaf = \mathbf{F}$, and evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0, 0, 0)$ to $(1, 2, 3)$.

6 [Foundational] • Target: The Gradient Vector

Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for the vector field $\mathbf{F} = \langle y,\ -x, z \rangle$ along the circular helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ for $0 \le t \le 2\pi$.



7 [Foundational] • Target: Directional Derivatives

Consider the vector field $\mathbf{F}(x, y) = \langle 2x + y, x - 3y \rangle$. What is the value of the vector field at the point $(3, -1)$?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

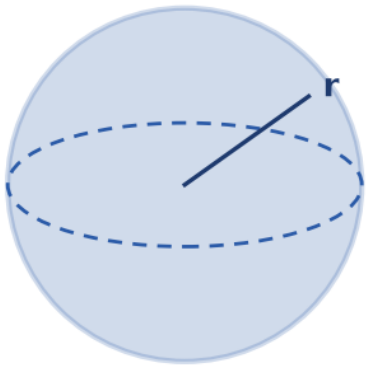
8 [Intermediate] • Target: Direction of Maximum Increase & Steepest Ascent

Maximize $f(x, y) = 5xy$ subject to $x + y = -10$. Find the maximum value.

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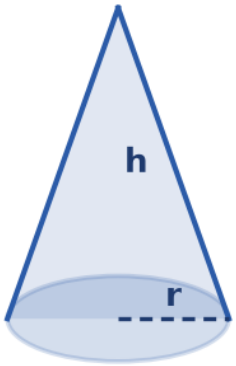
9 **[Intermediate]** • **Target: Multivariable Chain Rule**

Find the total mass and the z-coordinate of the center of mass of the upper solid hemisphere $x^2 + y^2 + z^2 \le R^2$, $z \ge 0$, whose density is proportional to the distance from the origin: $\delta(\rho, \phi, \theta) = k\rho$ for constant $k > 0$.



10 **[Intermediate]** • **Target: The Gradient Vector**

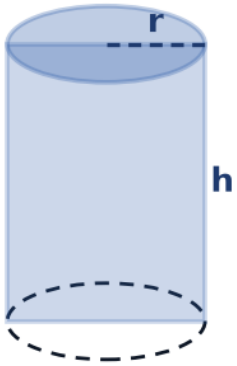
Compute the moment of inertia I_z about the z-axis of a solid homogeneous right circular cone of base radius R, height h, and uniform total mass M.



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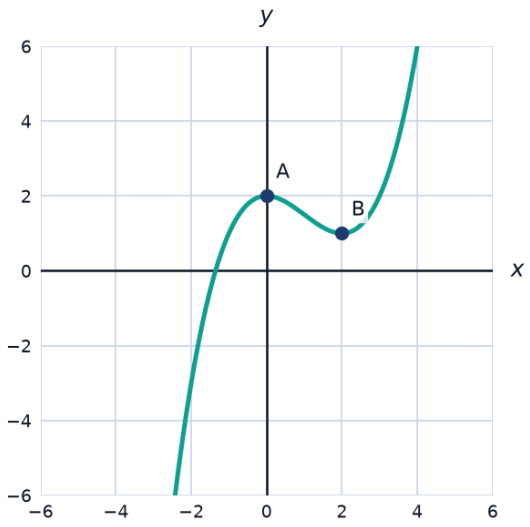
11 [Intermediate] • Target: Directional Derivatives

Find the volume and the z-coordinate of the centroid $\bar{z}$ of the solid region bounded below by the xy-plane $z = 0$ and above by the paraboloid $z = 4 - x^2 - y^2$, assuming uniform density $\delta_0 = 1$.



12 [Intermediate] • Target: Direction of Maximum Increase & Steepest Ascent

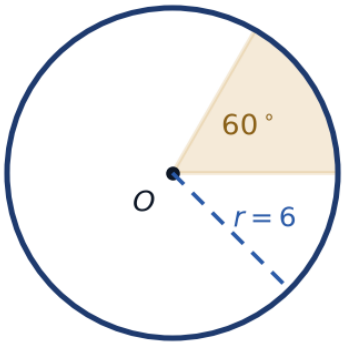
Evaluate the Cartesian triple integral $\iiint_B xyz^2\,dV$ over the rectangular box $B = [0, 1] \times [-1, 2] \times [0, 3]$.



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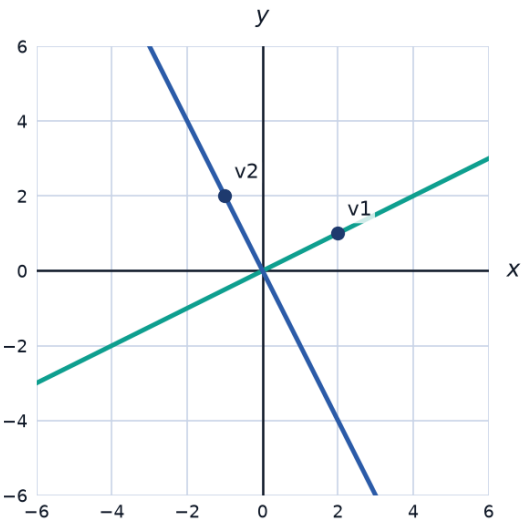
13 [Intermediate] • Target: Multivariable Chain Rule

Evaluate the double integral $\iint_R x\cos(x+y)\,dA$ over the rectangular region $R = [0, \pi] \times [0, \pi/2]$.



14 [Intermediate] • Target: The Gradient Vector

Classify the critical points $(0, 0)$ and $(1, 1)$ of $f(x, y) = x^3 + y^3 - 3xy$ using the Second Derivative Test and Hessian matrix.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Directional Derivatives • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Consider the solid region E bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by the sphere $x^2 + y^2 + z^2 = 16$. The volume of E is given by the triple integral $V = \iiint_E dV$. If E is expressed in spherical coordinates, the integral becomes $\int_\alpha^\beta \int_\gamma^\delta \int_{\rho_1}^{\rho_2} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$. Determine the value of α, the lower limit of integration for θ.

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16 [Challenging] • Target: Direction of Maximum Increase & Steepest Ascent • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Find the surface area of the paraboloid cap $z = x^2 + y^2$ that lies below the horizontal plane $z = 4$.

17 [Challenging] • Target: Multivariable Chain Rule • Cross-Lesson: 4.01 Double Integrals

Prove that the limit $\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2}$ does not exist by using the Two-Path Test, comparing straight-line approaches with parabolic approaches.

18 [Challenging] • Target: The Gradient Vector • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Consider the function $f(x, y) = x^3 + y^3 - 3axy$, where a is a positive constant. It is known that $f(x, y)$ has a critical point at (x_0, y_0) where $x_0 = 2$ and $y_0 = 2$. Find the value of a.

19 [Challenging] • Target: Directional Derivatives • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Consider the region R in the xy-plane defined by $0 \le x \le 1$ and $0 \le y \le x^2$. Evaluate the double integral $\iint_R y\,dA$.

20 [Challenging] • Target: Direction of Maximum Increase & Steepest Ascent • Cross-Lesson: 4.01 Double Integrals

A solid region E is defined by the inequalities $x^2 + y^2 \le 16$, $0 \le z \le 5$. The volume of this region can be computed using a triple integral. In cylindrical coordinates, the region E is described by $0 \le r \le R$, $0 \le \theta \le 2\pi$, and $0 \le z \le H$. What is the value of R?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multivariable Chain Rule

If $z = f(x, y)$ where $x = g(t)$ and $y = h(t)$, what is $\frac{dz}{dt}$?

a. $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$

b. $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dy}{dt} + \frac{\partial z}{\partial y}\frac{dx}{dt}$

c. $\frac{dz}{dt} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$

d. $\frac{dz}{dt} = \frac{dx}{dt}\frac{dy}{dt}$

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22 [Foundational] • Target: Multivariable Chain Rule

Given $z = x^2y$, $x = t^3$, and $y = t^2$, find $\frac{dz}{dt}$ at $t = 1$.

a. 8

b. 6

c. 5

d. 12

23 [Foundational] • Target: Multivariable Chain Rule

If $z = f(x, y)$ with $x = r\cos\theta$ and $y = r\sin\theta$, what is $\frac{\partial z}{\partial r}$?

a. $\frac{\partial z}{\partial x}\cos\theta + \frac{\partial z}{\partial y}\sin\theta$

b. $-r\frac{\partial z}{\partial x}\sin\theta + r\frac{\partial z}{\partial y}\cos\theta$

c. $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$

d. $\cos\theta + \sin\theta$

24 [Foundational] • Target: The Gradient Vector

What is the gradient vector $\nabla f(x, y)$ of a scalar function $f(x, y)$?

a. $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$

b. $\nabla f = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$

c. $\nabla f = \left[\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}\right]$

d. $\nabla f = \sqrt{f_x^2 + f_y^2}$

25 [Foundational] • Target: The Gradient Vector

Compute the gradient $\nabla f(2, -1)$ for $f(x, y) = 3x^2y + 2y^2$.

a. $[-12, 8]$

b. $[12, -8]$

c. $[-12, -4]$

d. $[6, 4]$

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26 [Foundational] • Target: The Gradient Vector

What is the geometric relationship between the gradient $\nabla f(x_0, y_0)$ and the level curve $f(x, y) = c$ passing through (x_0, y_0)?

a. $\nabla f(x_0, y_0)$ is perpendicular (orthogonal) to the tangent line of the level curve

b. $\nabla f(x_0, y_0)$ is parallel to the tangent line

c. $\nabla f(x_0, y_0)$ has zero dot product with the normal vector

d. $\nabla f(x_0, y_0)$ points along the curve

27 [Foundational] • Target: Directional Derivatives

What is the formula for the directional derivative $D_{\mathbf{u}}f$ in the direction of a unit vector $\mathbf{u}$?

a. $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$

b. $D_{\mathbf{u}}f = \nabla f \times \mathbf{u}$

c. $D_{\mathbf{u}}f = ||\nabla f|| ||\mathbf{u}||$

d. $D_{\mathbf{u}}f = \frac{\nabla f \cdot \mathbf{u}}{||\nabla f||}$

28 [Foundational] • Target: Directional Derivatives

If $\nabla f(1, 2) = [4, -3]$ and $\mathbf{u} = \left[\frac{3}{5}, \frac{4}{5}\right]$, find $D_{\mathbf{u}}f(1, 2)$.

a. 0

b. 5

c. $\frac{24}{5}$

d. -1

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Directional Derivatives

Why must $\mathbf{u}$ be normalized to a unit vector before computing $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$?

a. To ensure the derivative measures rate of change per unit of geometric distance

b. Because dot products cannot be computed with non-unit vectors

c. To avoid division by zero

d. Because the gradient has length 1

30 [Intermediate] • Target: Direction of Maximum Increase & Steepest Ascent

In which direction does a differentiable function f increase most rapidly at a point?

a. In the direction of the gradient vector ∇f

b. In the direction opposite to ∇f (i.e. $-\nabla f$)

c. Orthogonal to the gradient vector

d. Along the tangent to the level curve

31 [Intermediate] • Target: Direction of Maximum Increase & Steepest Ascent

What is the maximum value of the directional derivative $D_{\mathbf{u}}f$ at a point?

a. $||\nabla f||$

b. $||\nabla f||^2$

c. 0

d. 1

32 [Intermediate] • Target: Direction of Maximum Increase & Steepest Ascent

In which direction is the rate of change of f equal to zero?

a. In any direction tangent to the level curve (orthogonal to ∇f)

b. In the direction of $-\nabla f$

c. In the direction of ∇f

d. Nowhere, a differentiable function always changes

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33 [Intermediate] • Target: Multivariable Chain Rule

Let the scalar field be given by $f(x, y) = 3x^2 - 5y$. Compute the value of $f_x(2, -3)$.

a. 6

b. 12

c. -6

d. 15

34 [Intermediate] • Target: The Gradient Vector

For $f(x, y) = -9x - 40y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 41

b. 49

c. 40

d. 42

35 [Intermediate] • Target: Directional Derivatives

For $f(x, y) = 20x - 21y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 41

b. 29

c. 28

d. 30

36 [Intermediate] • Target: Direction of Maximum Increase & Steepest Ascent

For $f(x, y) = -15x - 8y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 23

b. 16

c. 18

d. 17

37 [Intermediate] • Target: Multivariable Chain Rule

For $f(x, y) = -7x - 24y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 31

b. 25

c. 24

d. 26

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The Gradient Vector • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

For $f(x, y) = -24x - 7y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 31

b. 24

c. 26

d. 25

39 [Challenging] • Target: Directional Derivatives • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

For $f(x, y) = -21x + 20y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 41

b. 29

c. 28

d. 30

40 [Challenging] • Target: Direction of Maximum Increase & Steepest Ascent • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 8x - 15y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 23

b. 16

c. 18

d. 17

41 [Challenging] • Target: Multivariable Chain Rule • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

For $f(x, y) = 24x - 7y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 31

b. 24

c. 26

d. 25

42 [Challenging] • Target: The Gradient Vector • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

For $f(x, y) = -15x + 8y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 23

b. 16

c. 18

d. 17

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43 [Challenging] • Target: Directional Derivatives • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = -6x - 8y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 10

b. 14

c. 9

d. 11

44 [Challenging] • Target: Direction of Maximum Increase & Steepest Ascent • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

For $f(x, y) = 12x - 5y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 13

b. 17

c. 12

d. 14

45 [Challenging] • Target: Multivariable Chain Rule • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

For $f(x, y) = -9x + 40y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

a. 49

b. 40

c. 41

d. 42

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